Shopping Trends: Purchase Behavior Analysis

From SQL extraction to business recommendations

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Analysis Context

Problem Statement: Why do promotions fail to increase revenue despite an average 25% discount?

$233K
Total Revenue

3,900
Transactions

$59.76
Average Basket

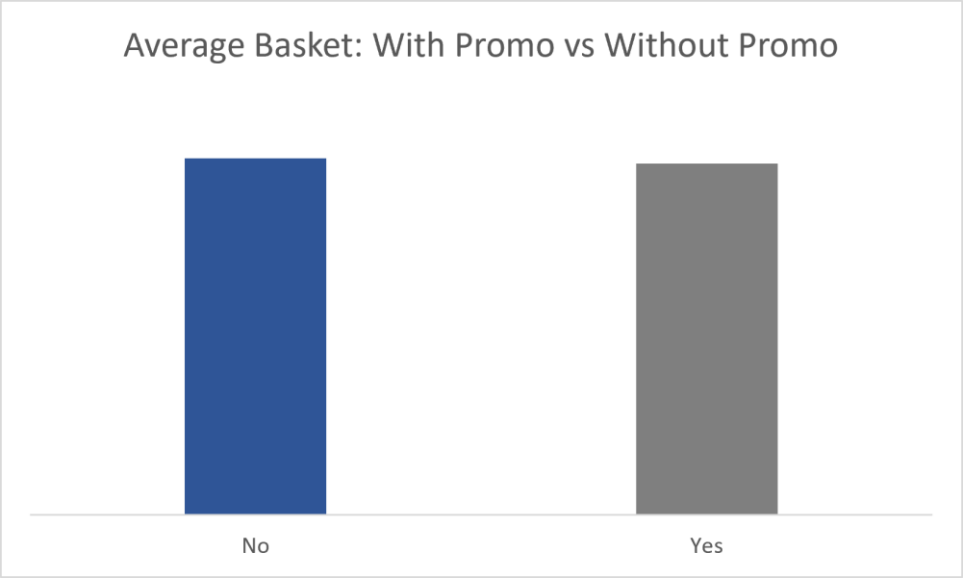
3.75/5
Average Rating

Promotions do not increase basket size

Observation: 43% of orders use a promo code. The average Basket with a promo (60.13).

Recommendation: Stop general promotions. Reserve them exclusively for new customer acquisition.

Estimated impact: +5 to 8% annual margin.

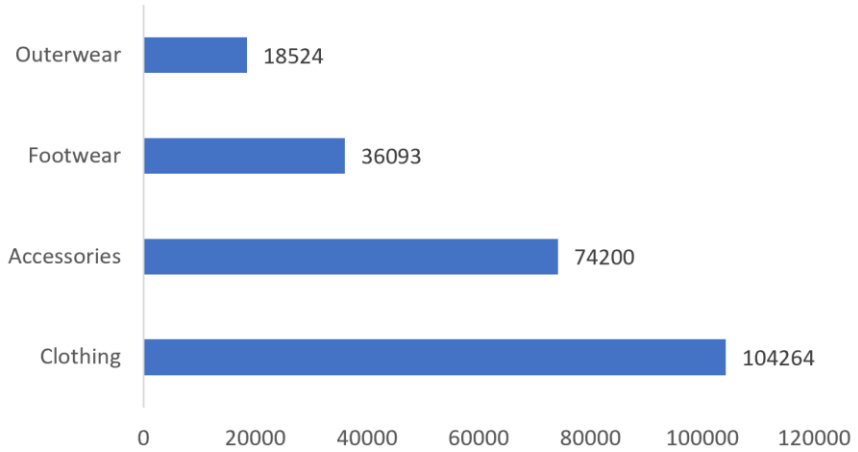


Revenue is concentrated in 2 categories

Observation: Clothing and Accessories together account for 76% of total revenue ($178,464 out of $233,000)

Recommendation: Secure stock and marketing budgets for these two categories first.

Estimated impact: A 10% drop annual margin would cost $10,460 in revenue.

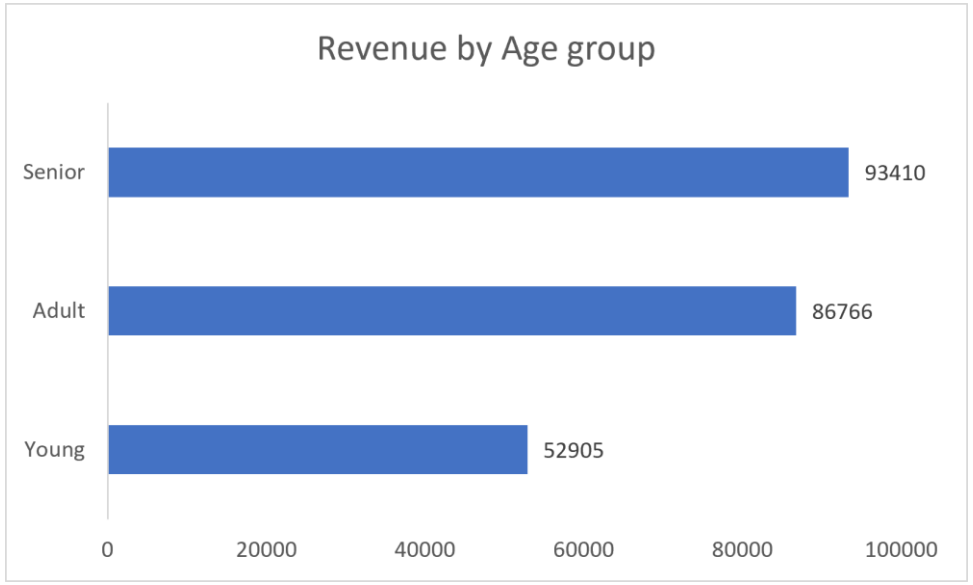


The Young Female segment is underexploited

Observation: Adult and Senior make up 77% of revenue. However, young women (young female) have the highest average basket in the dataset **($61.82**).

Recommendation: Launch a targeted premium acquisition campaign for young women.

Estimated impact: +$17,000 by doubling their purchase volume .



Priority Action Plan

Action	Impact	Priority	Timeline
Stop general promotions	+5 to 8% margin	High	Q1
Protect Clothing & Accessories	-$10K risk avoided	Medium	Q2
Target Young female segment	+$17K revenue	Medium	Q3

Estimated global Impact : +8 to 11% annual margin

Next Steps

- A/B test to remove general promotions (30 days)
- Deploy a Power BI dashboard for KPI tracking
- Customer satisfaction Audit (rating at 3.75/5)

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Methodology & Tools

- **Extraction & Analysis** : PostgreSQL (Aggregations, Window Functions), Power Query.
- **Visualization**: Excel (Pivot tables, conditional formatting)
- **Volume:** 3,900 transactions over 12 months.
- **Source:** Shopping Trends dataset